

Iris Flower Classification

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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = KNeighborsClassifier(n_neighbors=3)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", round(accuracy * 100, 2), "%")
```

```
sepal_length = float(input("Enter Sepal Length (cm): "))
sepal_width = float(input("Enter Sepal Width (cm): "))
petal_length = float(input("Enter Petal Length (cm): "))
petal_width = float(input("Enter Petal Width (cm): "))

prediction = model.predict([[sepal_length, sepal_width, petal_length,
petal_width]])

print("\nPredicted Iris Flower:", iris.target_names[prediction[0]])
```

Output:

Model Accuracy: 100.0 %

Enter Sepal Length (cm): 5.1

Enter Sepal Width (cm): 3.5

Enter Petal Length (cm): 1.4

Enter Petal Width (cm): 0.2

Predicted Iris Flower: setosa